Low Power TG Full Adder Design Using CMOS Nano Technology

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Abstract— **Full adders is the basic building block of ALU and ALU is a basic functioning unit of the microprocessors and DSP. In the world of technology it has become essential to develop various new design methodologies to reduce the power and area consumption. In this paper transmission gates have been used to develop the proposed full adder using 6 transistors XOR gates. The carry logic has been efficiently implemented using 2x1 MUX to reduce transistor count. The reduction in Transistor count results in improved area and power consumption. The proposed full adder has been designed using 27 and 18 transistors using 90 nm technologies. The developed adder with 18 transistors has shown an improvement of 6.624% in power and 31.765% in area so as to implement adder efficiently in digital signal processors.**

Keywords- **DSCH, full adder, transmission gate (TG) very Large-scale integration (VLSI).**

I. INTRODUCTION

In VLSI design methodologies power minimization is one of the primary concerns because a long battery life is required for mobiles and portable devices. Power dissipation is increasing due increasing rate of transistors on a single chip. Adder is one of the most critical components of a processor, as it is used in the arithmetic logic unit (ALU), in the floating-point unit, and for address generation in case of cache or memory access [1]. Increasing demand for mobile electronic devices such as cellular phones, PDA's, and laptop computers requires the use of power efficient VLSI circuits. The Full Adder is a combinational circuit with three inputs, i.e. Bit1(A), Bit2(B), and Carry in(C) and two outputs i.e. Sum and carryout (Carry). The Boolean expressions for some and carry is given bellow:

$$\left\{ \begin{array}{l} sum = A \oplus B \oplus C \\ Carry = AB + BC + CA \end{array} \right\} \quad (1)$$

Where A, B, C are the inputs [2].

The power consumption of a CMOS digital circuit can be represented as

$$P = fCVdd^2 + fI_{short}Vdd + I_{leak}Vdd \quad (2)$$

Where f is the clock frequency, C is the average switched Capacitance per clock cycle, VDD is the supply voltage, I short is the short circuit current, and I leak is the off current . In a well-optimized low power VLSI circuit, the 1st term of Equation (2) is by far the dominant. The stand-by power consumption is accounted for by the 3rd term. Using a lower VDD is an effective way to reduce them dynamic power consumption since the 1st term is proportional to the square of VDD [3].

Transmission gate or pass gate act as a simple switch circuit and present a new class of logic circuits that used pass transistors or TG as a basic building block. As shown in Fig. 1, the CMOS transmission gate consists of one PMOS and one NMOS transistor, connected in parallel. The gate voltages applied to these two transistors as complementary signals. As such, the CMOS TG operates as a bidirectional switch between the nodes A and B which is controlled by signal C. If the control signal C is equal to *VDD,* then both transistors are turned on and provide a low-resistance current path between the nodes A and B. If, on the other hand, the control signal C is low, then both transistors will be off, and the path between the A and B will be an open circuit. Fig 1 also shows three other commonly used symbolic representations of the TG.

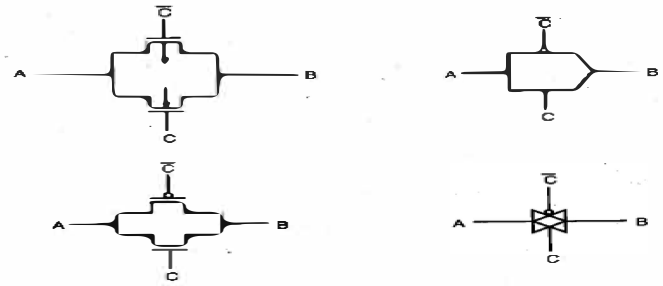


Fig 1-Symbolic representations of the transmission gate [4]

II. PREVIOUS WORKS

Various full adder circuits have been presented in literature [5] - [8]. 28 transistors conventional full adder is given in Fig.2 [5].In this adder design PMOS network is same as NMOS network.

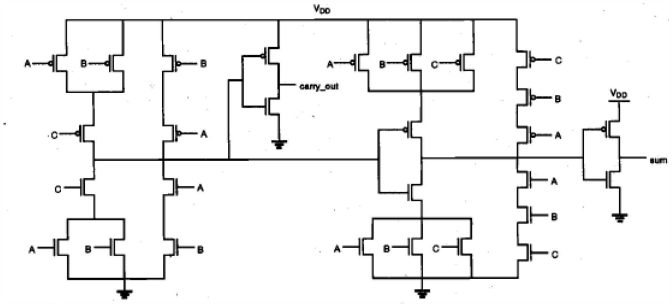


Fig.2 -The conventional design of CMOS Full Adder [5]

26 transistors transmission gate full adder is shown in Fig-2 [6] .Schematic of this circuit is simple as compared to previous one. This full adder consists less transistors as compared to previous one.

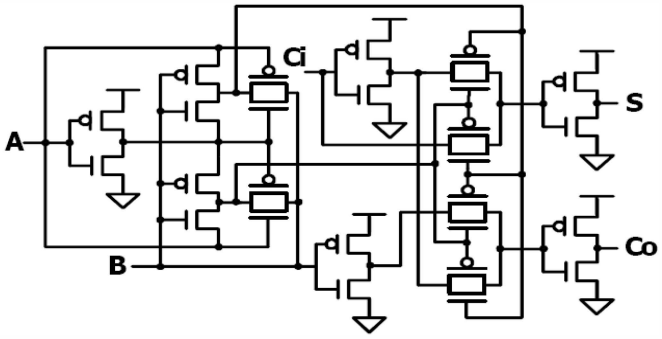


Fig.3 –TG Full Adder by 26 transistors [6]

20 transistors full adder by using pass transistor logic is shown in Fig-4 [7]. Although the transistor count is less as compared to previous one but the circuit consume more power and give the slower speed due to threshold loss problem.

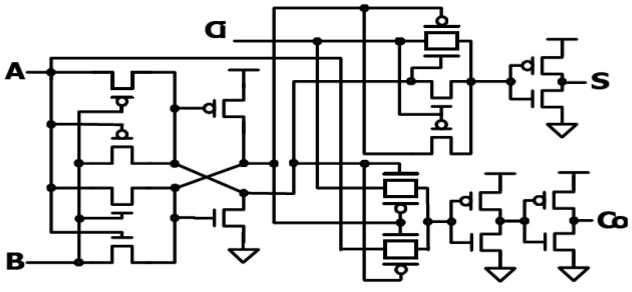


Fig 4- Pass transistor based Full Adder by 20 transistors [7]

A hybrid full adder by using pass transistor logic and static CMOS logic is shown in Fig-5 [8].Both sum and carry module needs 12 transistors respectively. So the total transistors needed for the design are 24 which are greater than the previous one. The adder circuit has been designed by 3 input XOR circuit.

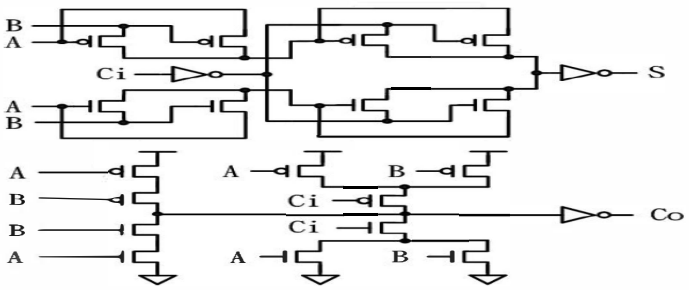


Fig 5- Pass transistor based Full Adder by 20 transistors [8]

III. TRANSISTOR COUNT OPTIMIZATION XOR CIRCUIT

CMOS transmission gate Figure 6 shows an eight-transistor implementation of the logic XOR function, using two CMOS TGs and two CMOS inverters.

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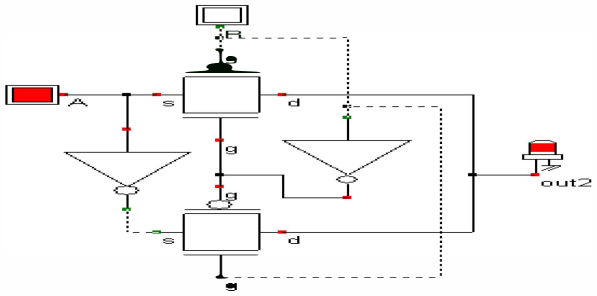


Fig-6 Eight-transistor CMOS TG implementation of the XOR function.

The same function can also be implemented using only six transistors, as shown in Fig. 7

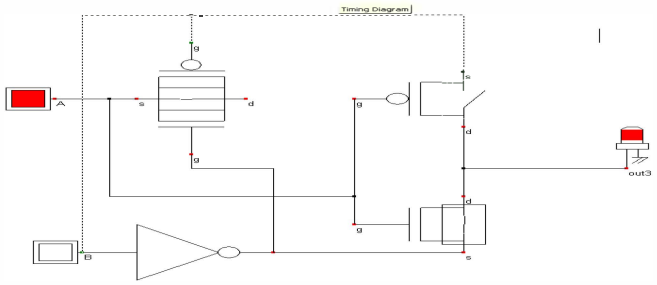


Fig.7- Six transistors CMOS TG implementation of the XOR

Comparisons of area and power of these two design is shown in TABLE-1

TABLE I. COMPARISON BETWEEN 8 AND 6 TRANSISTOR TG XOR

Comparison	***8 transistor TG XOR***	***6 transistor TG XOR***
Power Consume (μW)	9.497	2.63
Area (μm^2)	75	65.5

IV. SCHEMATIC DESIGN OF TG FULL ADDER

A. *TG full adder block diagram by using 6T XOR*

TG Full adder design of full adder by using 27 transistors is shown in Fig 5. This full adder circuit is made by using XOR circuit shown in Fig-8. Area and power consumed by this circuit is less as compared to full adder circuit made by XOR circuit in Fig -7.

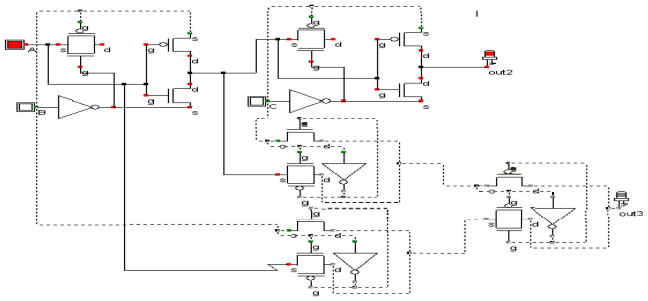


Fig 8 -TG Full adder design of Full Adder by using 27 transistors

B. *Proposed TG full adder by using 2x1 MUX*

The TG full adder circuit is implemented in MICROWIND by building following two modules:

A. One 2:1 MUX using 6 transistor

B. Two 2 input XOR gate using 6 transistors

Fig 9 shows a two-input multiplexor circuit consisting of two CMOS transmission gates. The operation of the multiplexor can be understood quite easily: If the control input S is logic-high, then the bottom TG will conduct and the output will be equal to the input B. If the control signal is low, the bottom TG will turn off and the top TG will connect the input A to the output node.

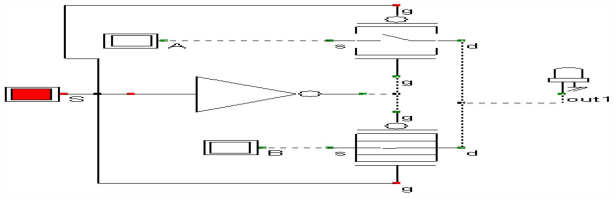


Fig 9- Two-input multiplexor circuit implemented using two TGs.

Optimized Architecture of TG Full Adder Circuit with 2x1 is shown in Fig-10. TG full adder using MUX in above Fig is shown in Fig-11 which consists only 18T.

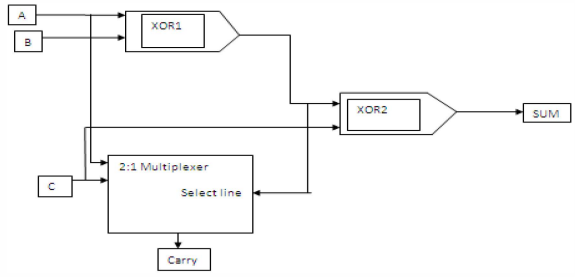


Fig 10- Conceptual diagram of TG Full Adder Circuit with 2x1

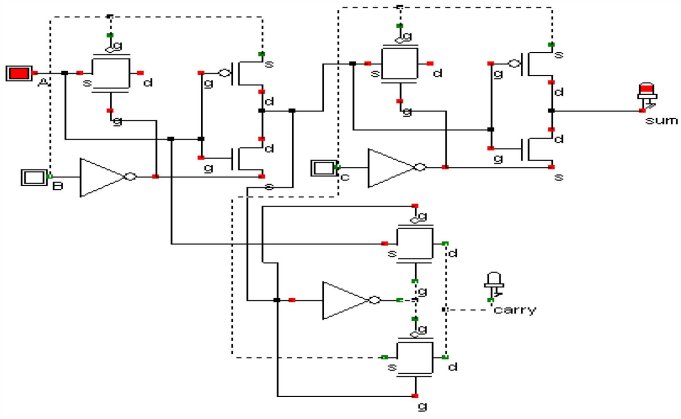


Fig 11 -TG Full Adder by using 2x1 MUX

V. SIMULATION AND LAYOUT

Reduction of power consumption provides a great improvement to an adder circuit. Power consumption issues can lead to over consumption of resources when devices are cascaded. This reduction in power would come at the expense of overall speed and increased delay.

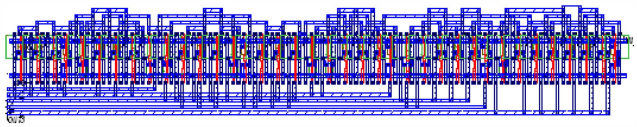


Fig 12- Layout of 27 transistor Full Adder

This section presents the layout and analog simulation of various designs. These designs are simulated using Microwind DSCH 3.1. The simulation is performed using schematic entry and its functionality is verified in DSCH. After verification VERILOG file is generated. VERILOG is compiled to get physical layout of schematic designs.

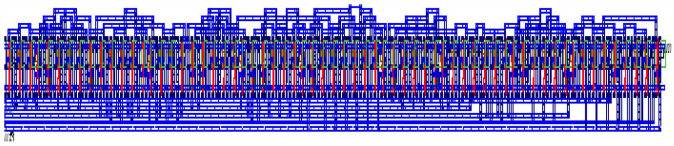


Fig-13 Layout of 18T Full Adder

Using layout of circuit values like power, resistance, capacitance, node voltage and current can be estimated .

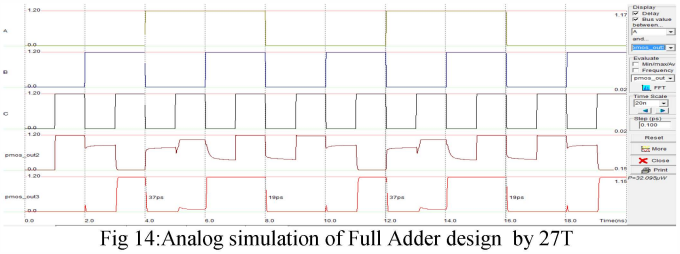


Fig 14:Analog simulation of Full Adder design by 27T

Layout and analog simulation of various designs are shown in Figs. Layout shows the various metal layers and their interconnection through via.Fig. 12 and 13 shows the layout and 14 and 15 shows the analog simulation of various designs.

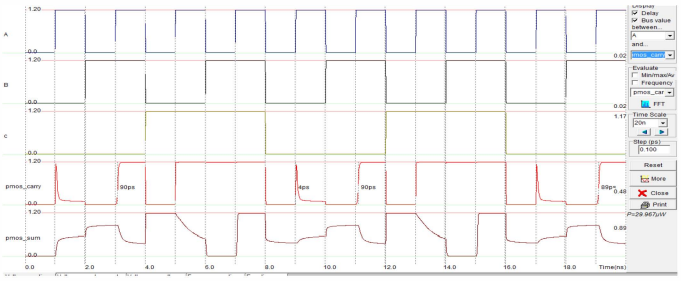


Fig 15:Analog simulation of Full Adder design by 18T

VI. EXPERIMENTALDISCRIPTION AND RESULTS

On the basis of MICROWIND designing tool we make the performance comparison between of area and power of CMOS full adder, TG full adder by 27T and TG full adder design with 2x1 MUX which consist 18 transistors. Full adder design by TG full adder 27T and TG full adder design with 18T have layout width and height of 44.5µm/7.4 µm and 32.1 µm /7 µm respectively

TABLE 2. COMPARATIVE STUDY OF AREA AND POWER OF VARIOUS TG FULL ADDER DESIGNS

Design	***Power (µW)***	***Area (µm²)***	***Transistor count***
TG FA/18	29.967	224.7	18
TG FA/27	32.093	329.3	27

Power consumed by each design is 509 µW, 32.093µW, 29.967 µW respectively. Power consumed by 18T full adder is 2.126 µW and area consumed is also 104.6 µm² less then 27T full adder.

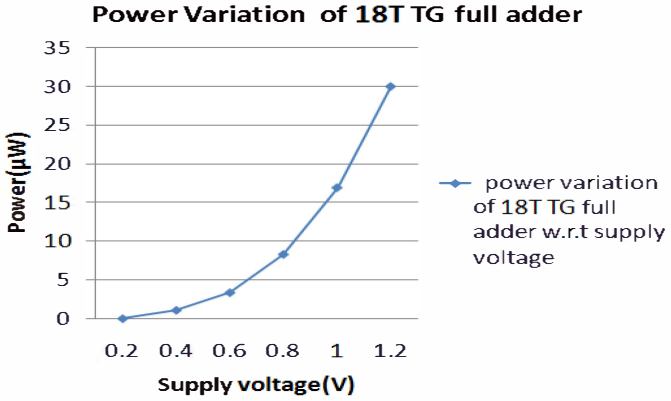


Fig 16 :Power variation of 18T Full Adder with Supply Voltage

Power variation of full adder designs with the supply voltage is shown in Fig 16, 17 respectively. At 1.2V input supply power consumed by the 27T full adder is 32.095µW whereas at the same input supply voltage 18T full adder consumes power equal to 29.967 µW. Power consumption is calculated at the 27ºC temperature.

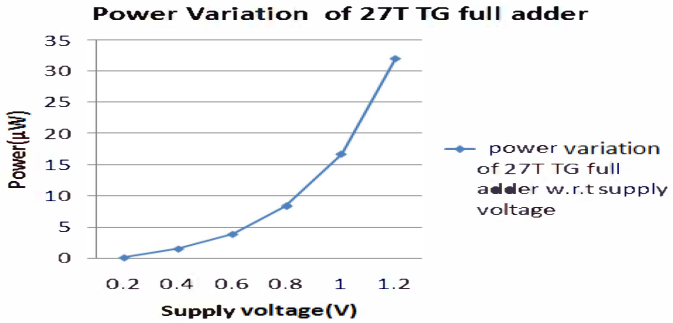


Fig 17: Power variation of 27T Full Adder with Supply Voltage

VII. CONCLUSION

This paper presents an area and power efficient technique to design a full adder, using TG and multiplexer in order to reduce transistor count. Most of the conventional CMOS adders have been designed using 28 transistors which are very high. This increased number of transistors results in high power consumption. To overcome this existing problem, the proposed full adder has been designed using less number of transistors to improve the power and area simultaneously. Two full adders have been designed using 27 and 18 transistors. The full adder using 27 transistors consumes power and area respectively 32.093µW and 329.3µm^2.Full adder using 18 transistors consumes power and area respectively 29.967µW and 224.7µm^2.All the results are simulated using MICROWIND3.1.

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